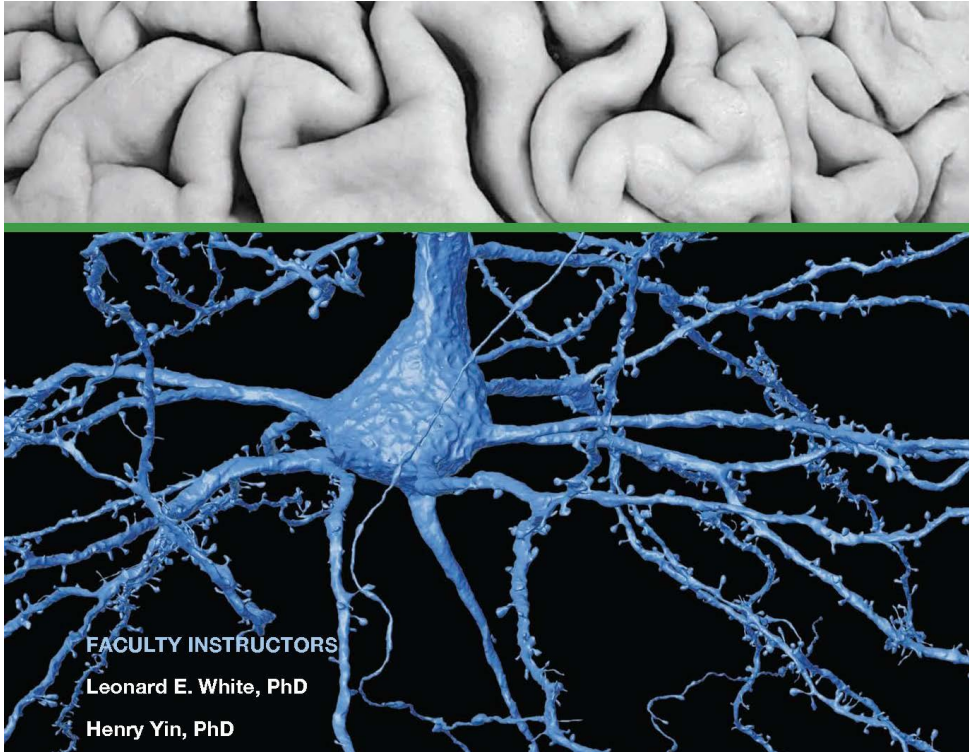


NEUROSCI 206L Introduction to Systems Neuroscience



Auditory System

Leonard E. White, PhD
Associate Professor in Neurology

KEY CONCEPTS

- 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds.
- 10.2 The structures of the ear filter the frequencies of sound and transmit vibrations from air to fluid.
- 10.3 Sound waves are transduced into neural signals by cochlear hair cells.
- 10.4 Transduction is controlled by active mechanisms involving middle ear muscles and cochlear outer hair cells.
- 10.5 Auditory pathways involve signals traveling bilaterally and in both feedforward and feedback directions.
- 10.6 Auditory perception involves the synthesis of multiple aspects of sound.
- 10.7 Neural codes for sound frequency are based on resonance and synchrony.
- 10.8 Sounds are localized based on frequency, relative loudness, and relative time of arrival at the ears.
- 10.9 Hearing coordinates with vision via numerous signal transformations and robust intersensory cross-talk.

Introduction to the Auditory System

- *the auditory system transduces sound waves into distinct patterns of neural activity that are then integrated with other sensations and motivations to guide behavior*

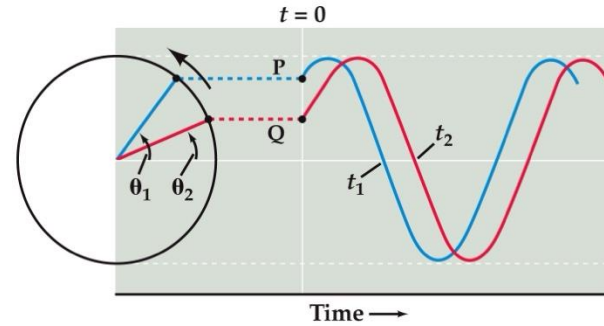
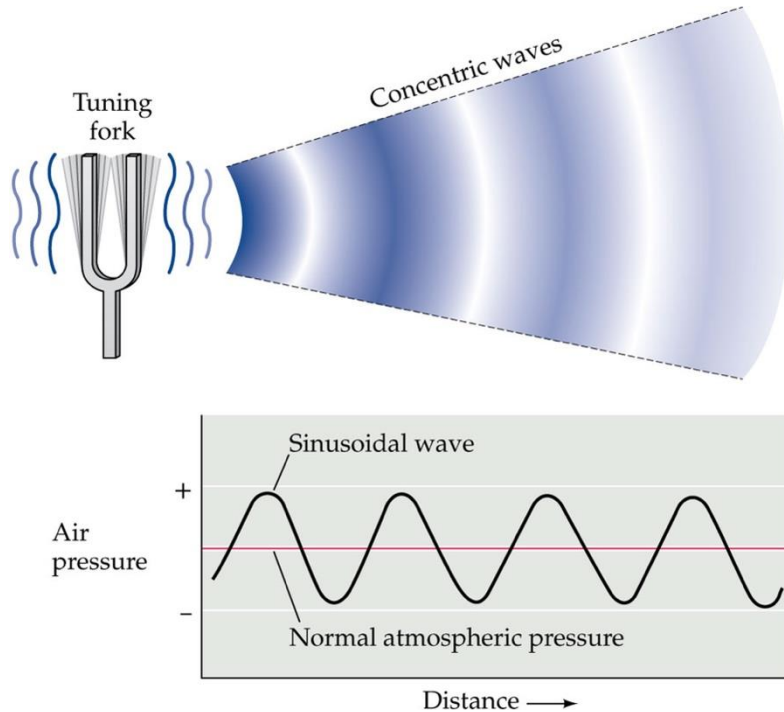
Introduction to the Auditory System

- sound waves are collected and amplified by physical structures in the external and middle ear for transfer to neural elements in the inner ear
- the biomechanical properties of the inner ear decompose complex sound waves into sinusoidal components, so that component frequencies, their amplitude, and their phase can be encoded in the firing of the receptor cells
- in the inner ear and throughout central processing stations, tonotopy is preserved; i.e., the systematic representation of sound frequency
- in the brainstem, the auditory information is first processed and divided into several parallel pathways, some of which compare the signals derived from the two ears, a process that allows for the localization of sounds in space

Introduction to the Auditory System

- in addition to local processing of auditory signals, brainstem centers relay information to the midbrain, which in turn projects to the auditory thalamus
- the auditory cortex then receives input from the thalamus and processes more complex aspects of sounds, such as those related to human speech
- at each stage of central auditory processing, there are extensive connections from “higher” centers to antecedent processing centers, and well as connections that integrate signals arriving at the two ears
- like all other sensory systems, hearing employs a variety of active mechanisms by which the brain can shape the incoming signals

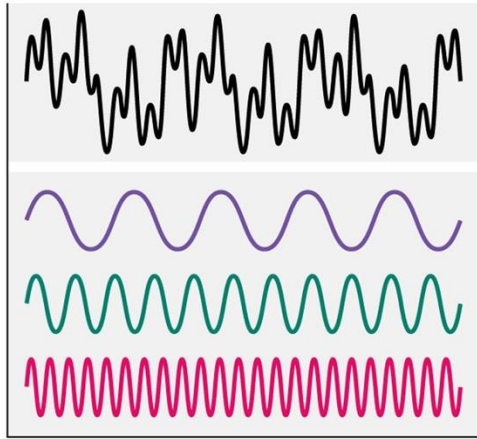
CONCEPT 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds



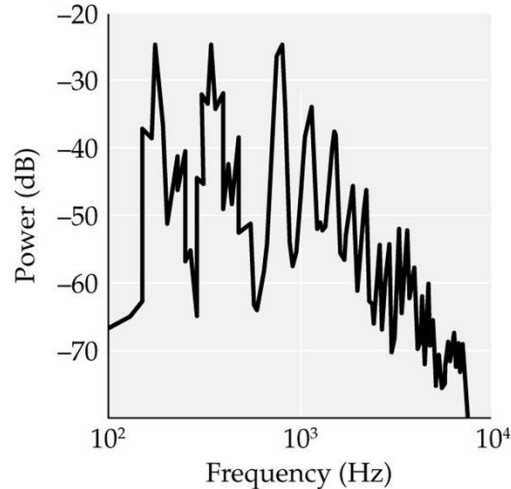
NEUROSCIENCE 5e, Figure 13.2
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CONCEPT 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds

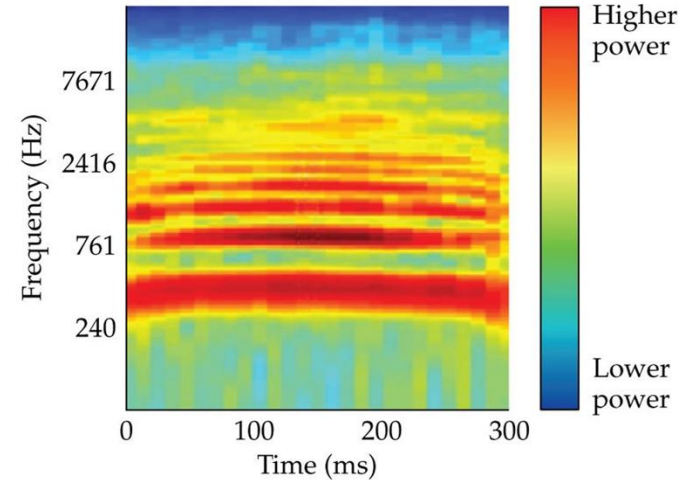
(A) Fourier decomposition



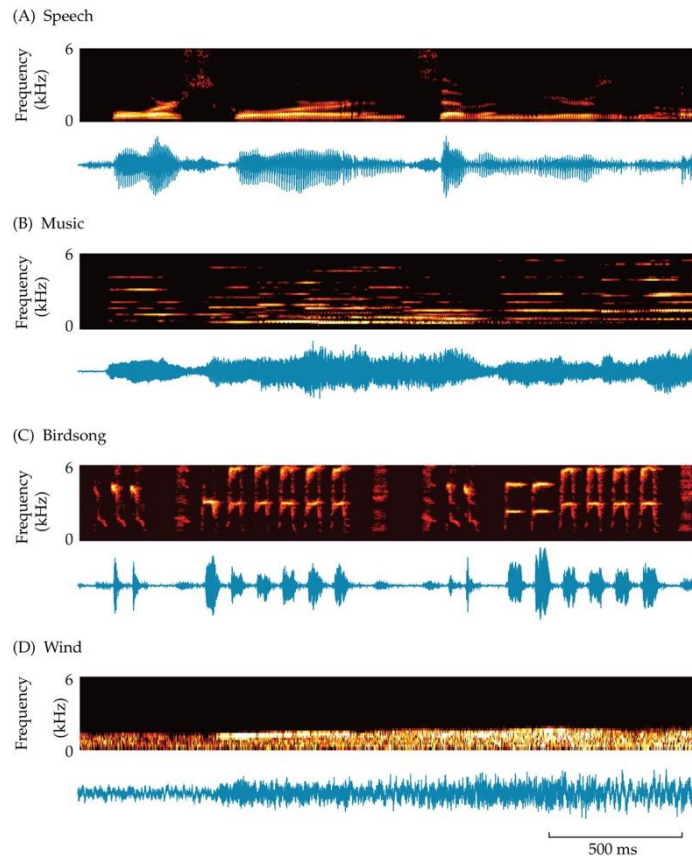
(B) Power spectrum



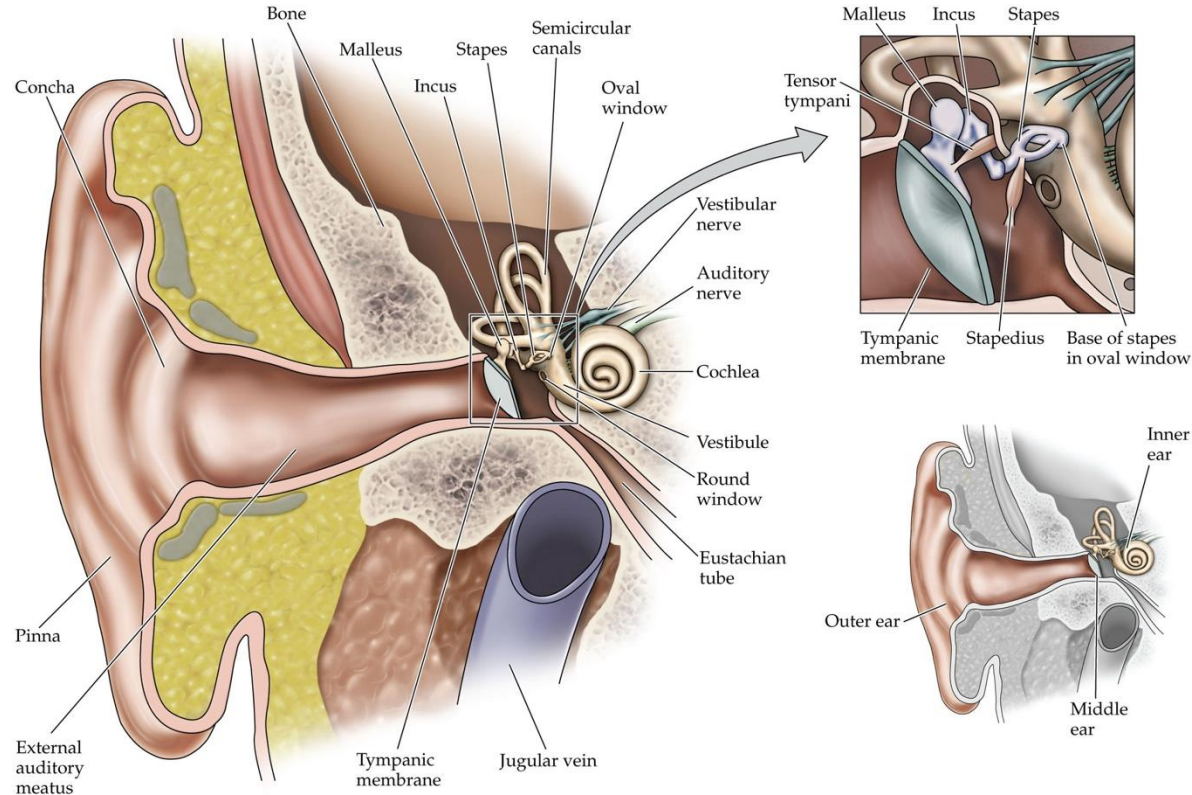
(C) Spectrogram



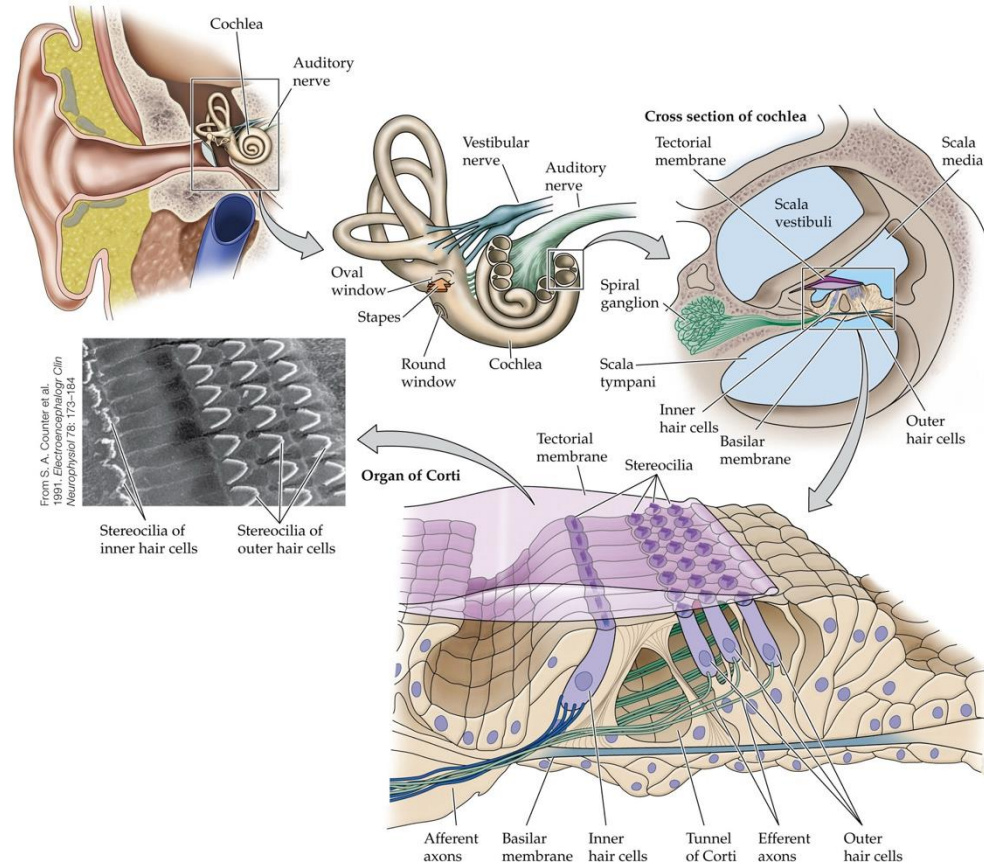
CONCEPT 10.1 Sound is a pressure wave composed of different frequencies important for speech, music, and other natural sounds



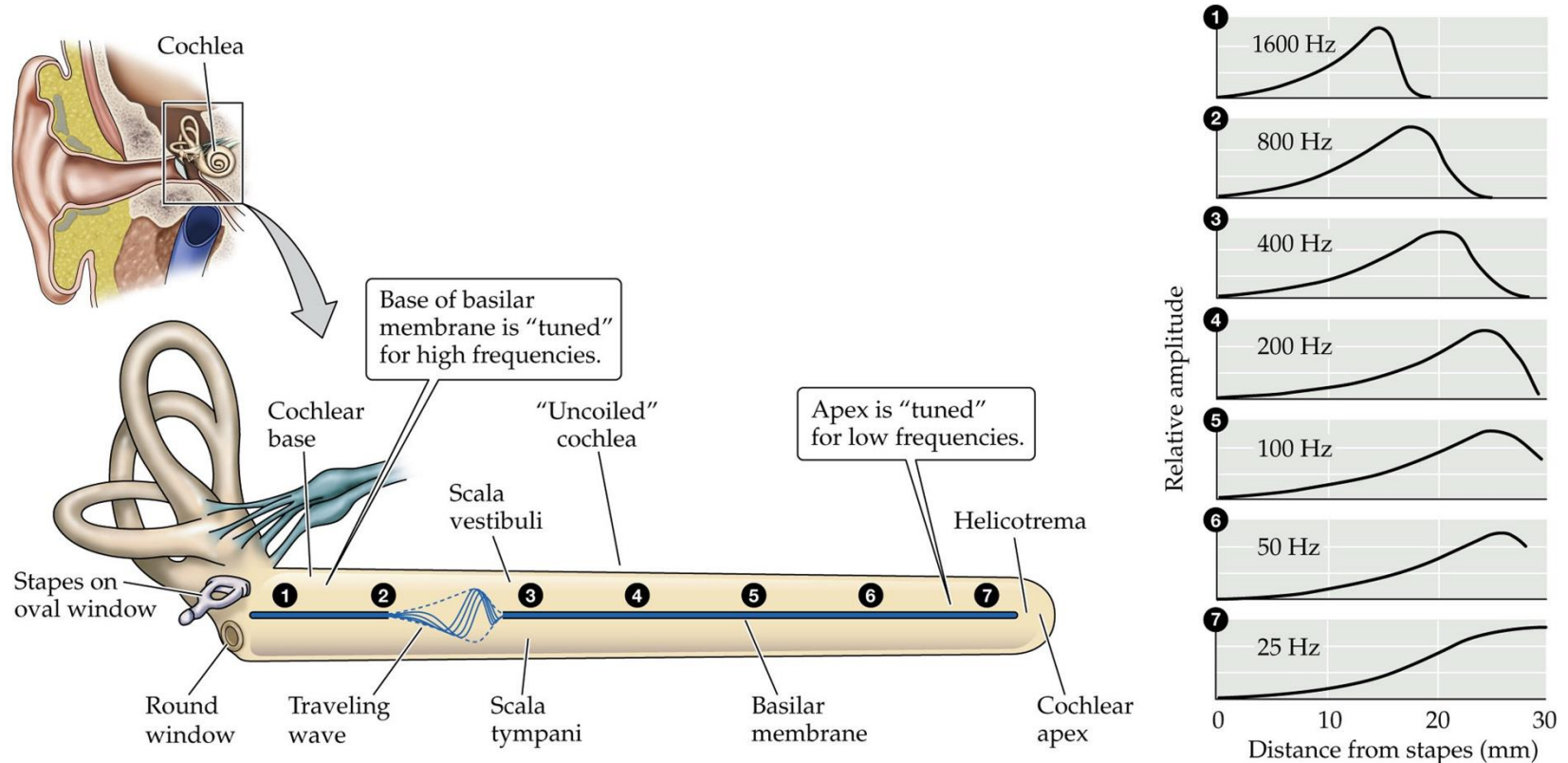
CONCEPT 10.2 The structures of the ear filter the frequencies of sound and transmit vibrations from air to fluid



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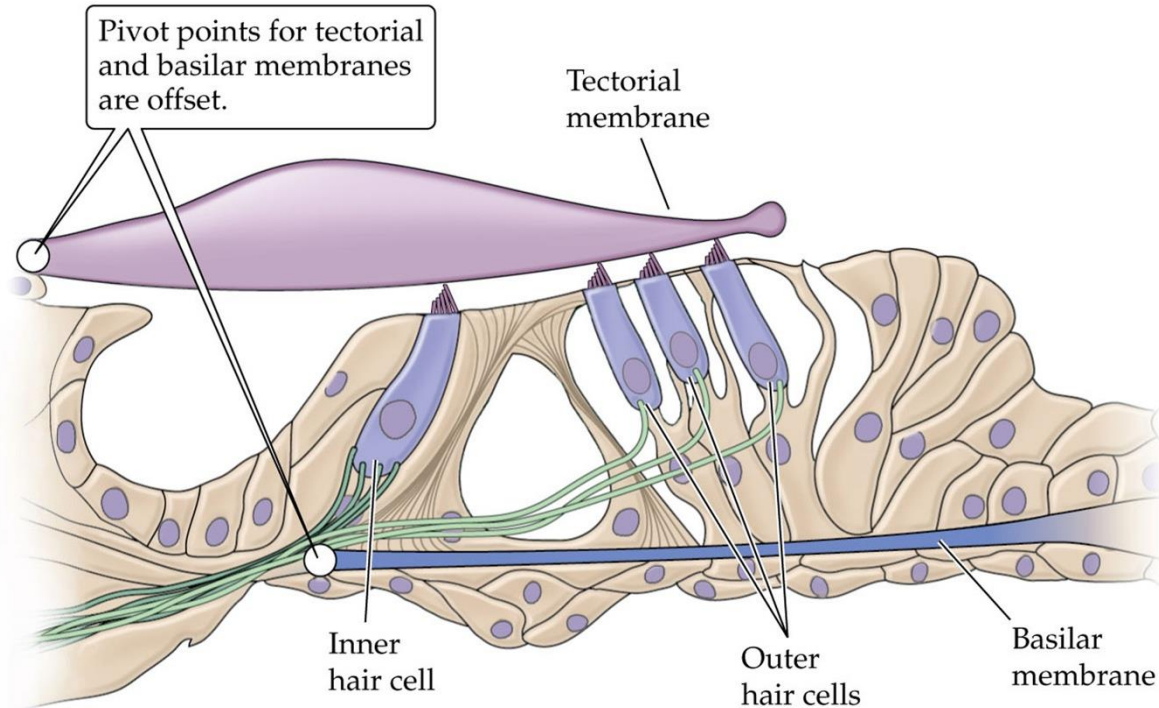
Sensory receptors: hair cells



Dr. Goran Bredberg/Science Photo Library

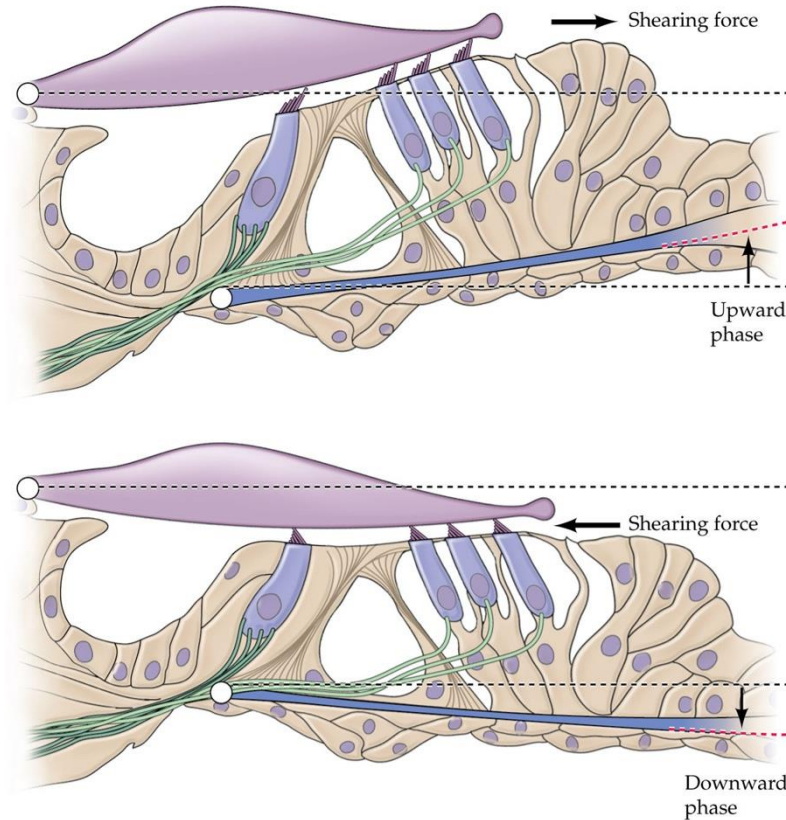
CONCEPT 10.3 Sound waves are transduced into neural signals by cochlear hair cells

(A) Resting position

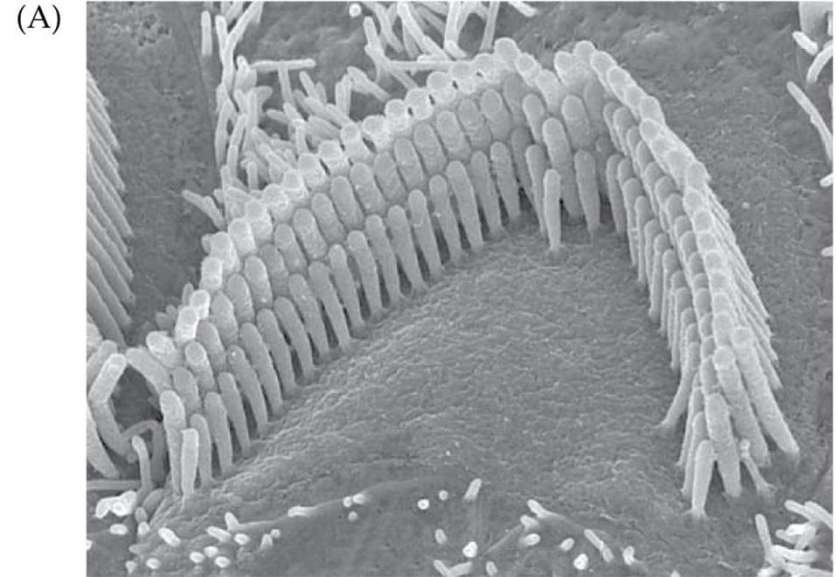
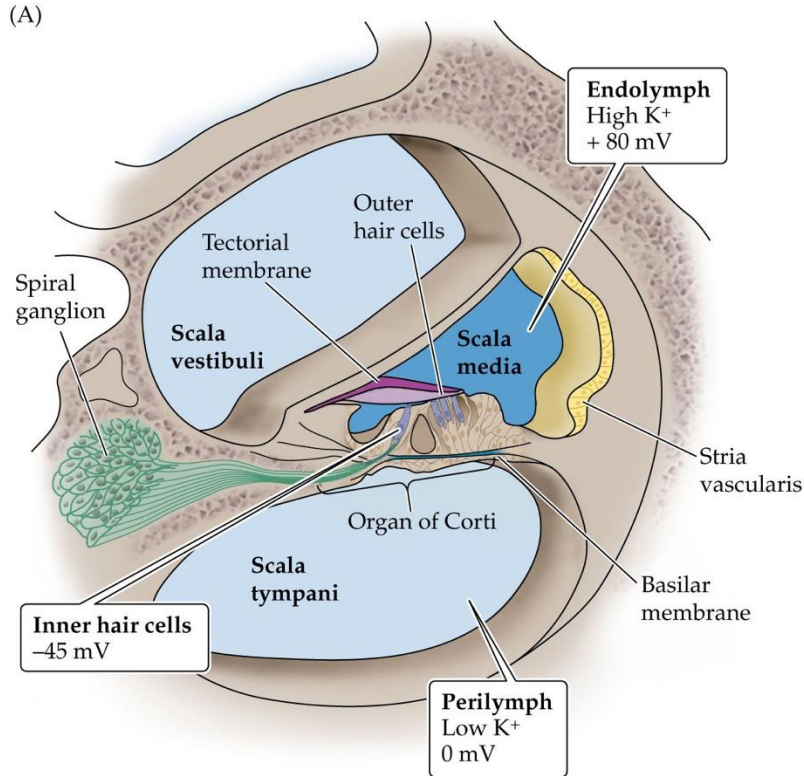


Biomechanics of the sensory transduction

(B) Sound-induced vibration

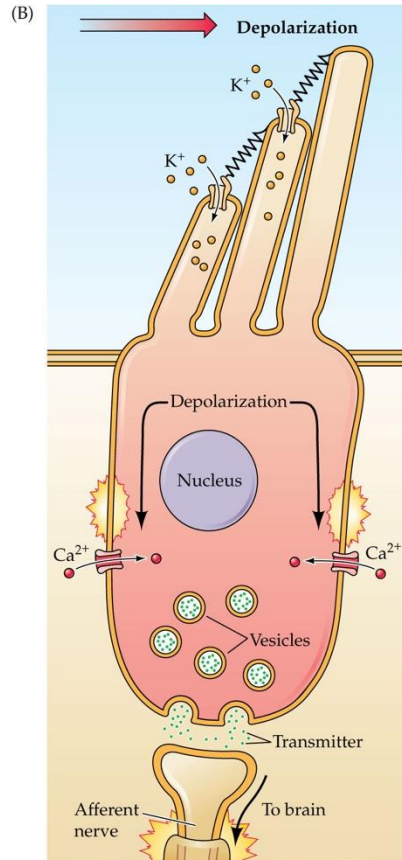


Neurophysiology of sensory transduction

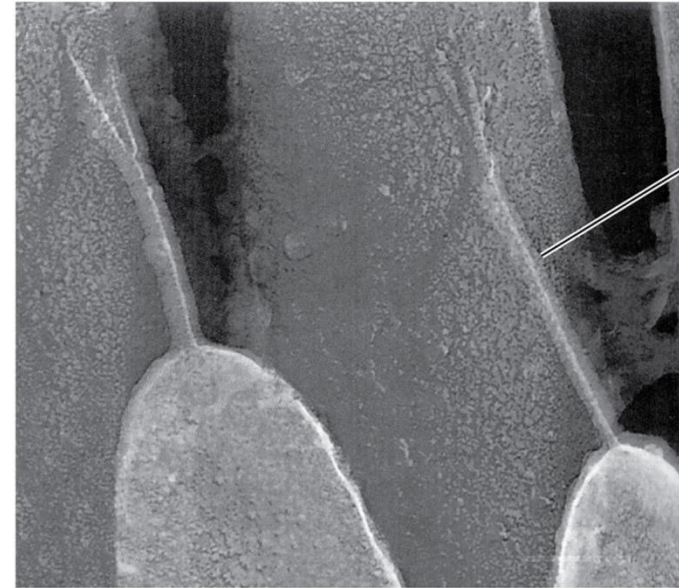


Courtesy of David Furness and
Carole Hackney, Keele University, UK

Depolarization by *influx* of K^+



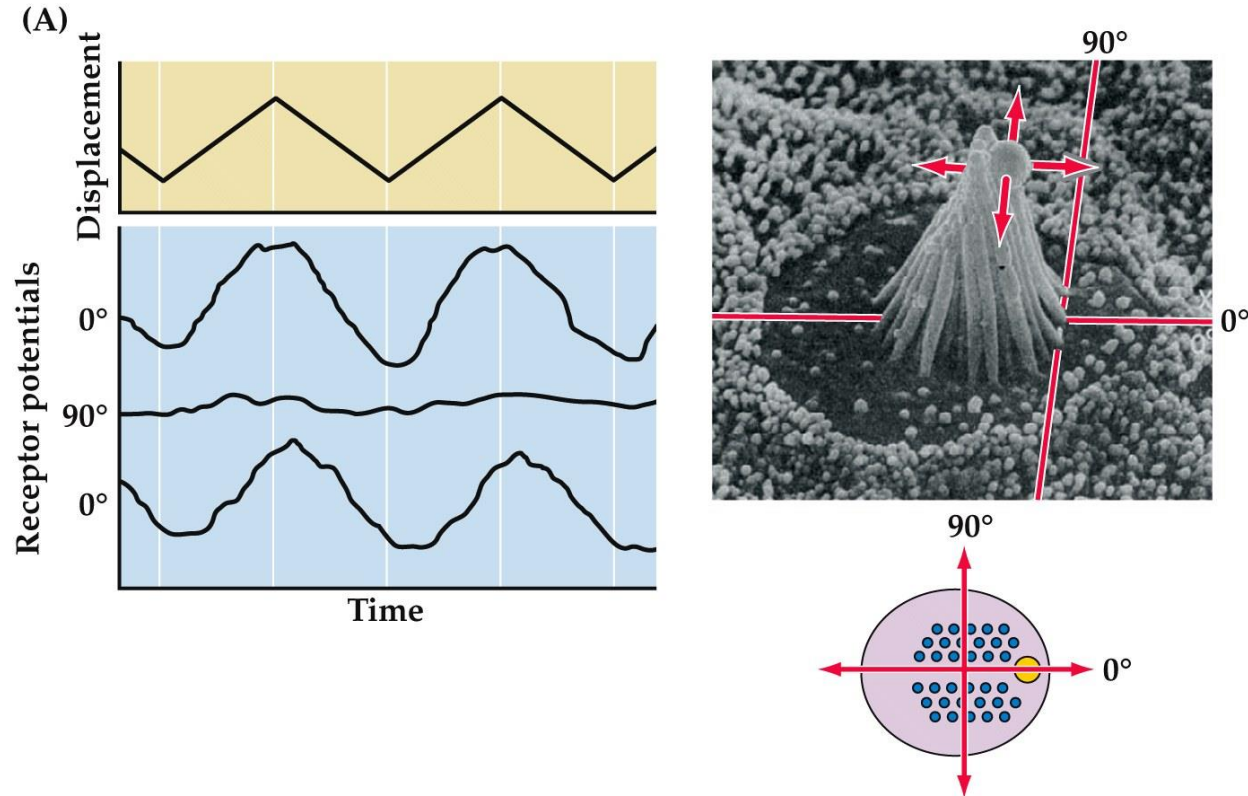
(B)



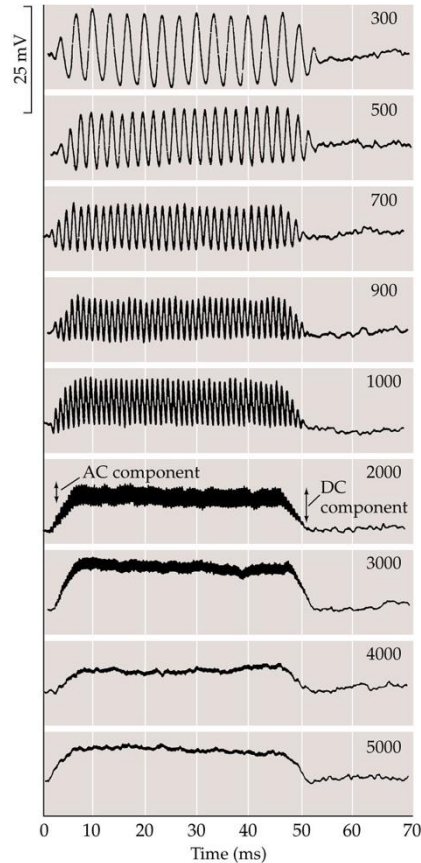
Tip link

From B. Kachar et al., 2000.
Proc Natl Acad Sci USA 97:
13336–13341. © 2000 National
Academy of Sciences, USA

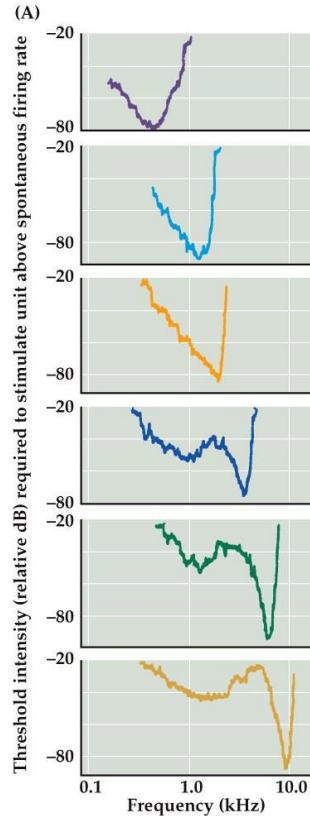
Stereocilium deflection determines membrane potential



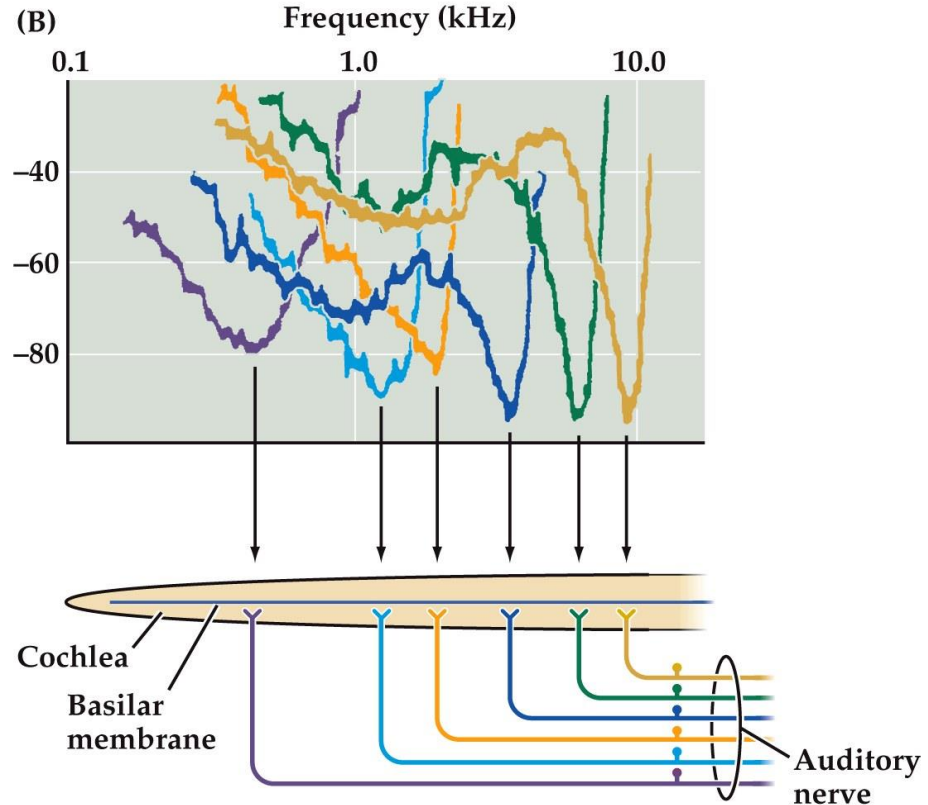
Frequency-dependent receptor potentials in inner hair cells



Labeled-line coding in CN VIII

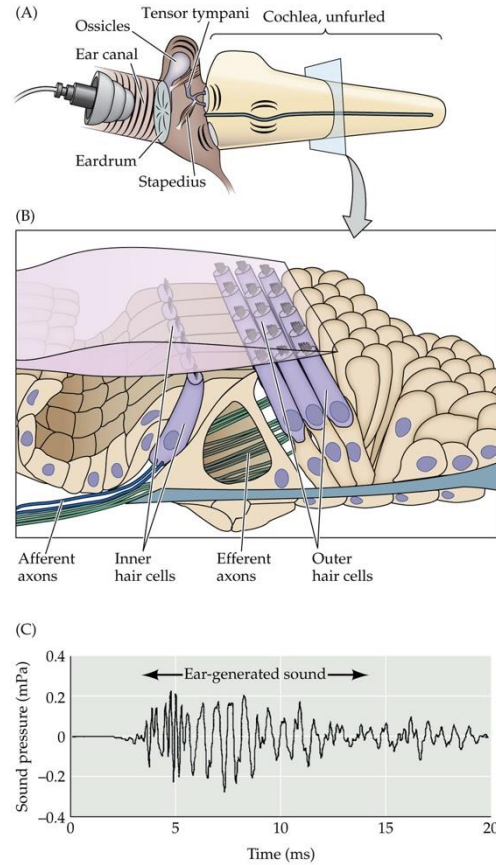


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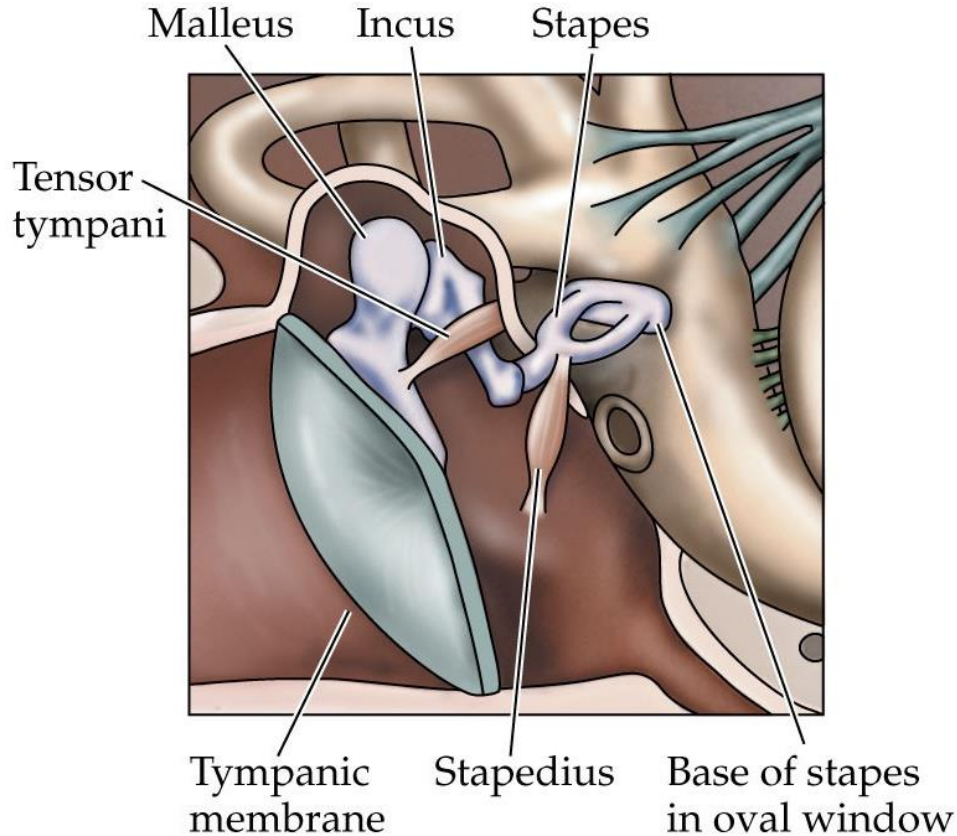


NEUROSCIENCE 5e, Figure 13.11 (Part 2)
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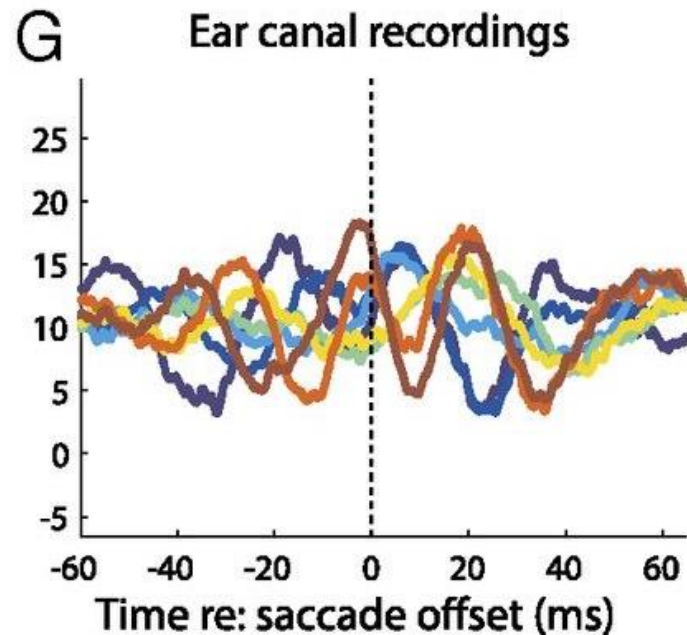
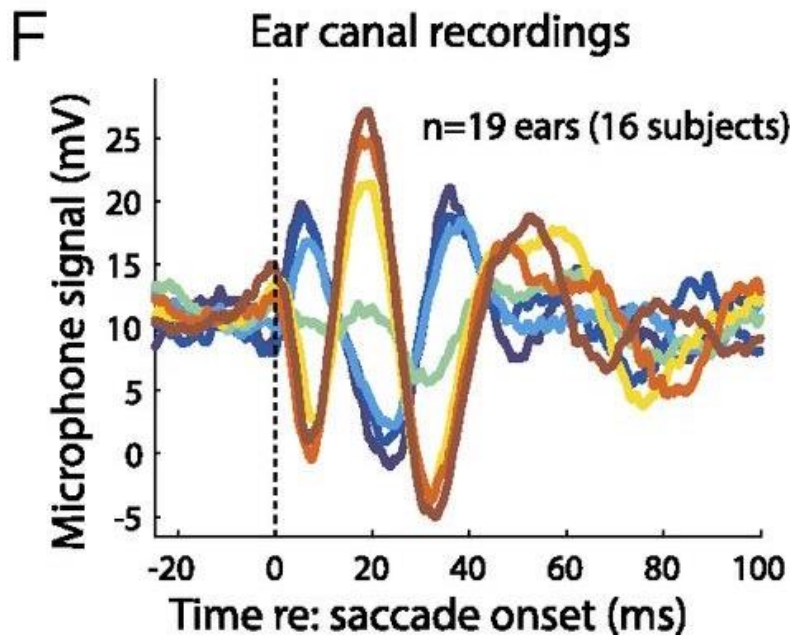
CONCEPT 10.4 Transduction is controlled by active mechanisms involving middle ear muscles and cochlear outer hair cells



Middle ear muscles dampen the action of the ossicles



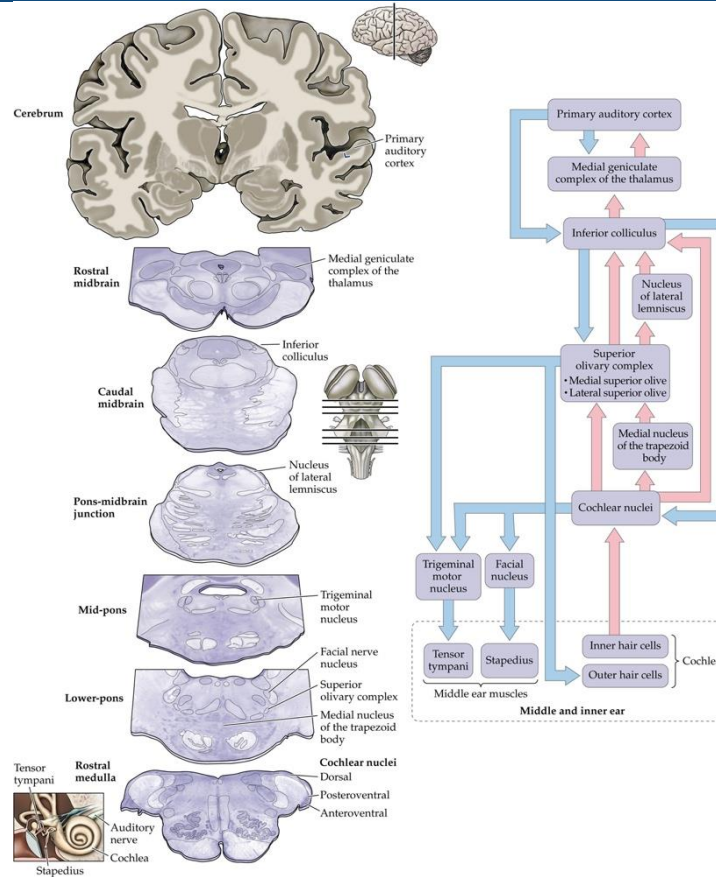
Active mechanisms facilitate the selective acquisition of energy from the environment



[from Figure 2 of K.G. Gruters, D.L.K. Murphy, C.D. Jenson, D.W. Smith, C.A. Shera, & J.M. Groh, The eardrums move when the eyes move: A multisensory effect on the mechanics of hearing. *Proc. Natl. Acad. Sci. U.S.A.* 115 (6) E1309-E1318]

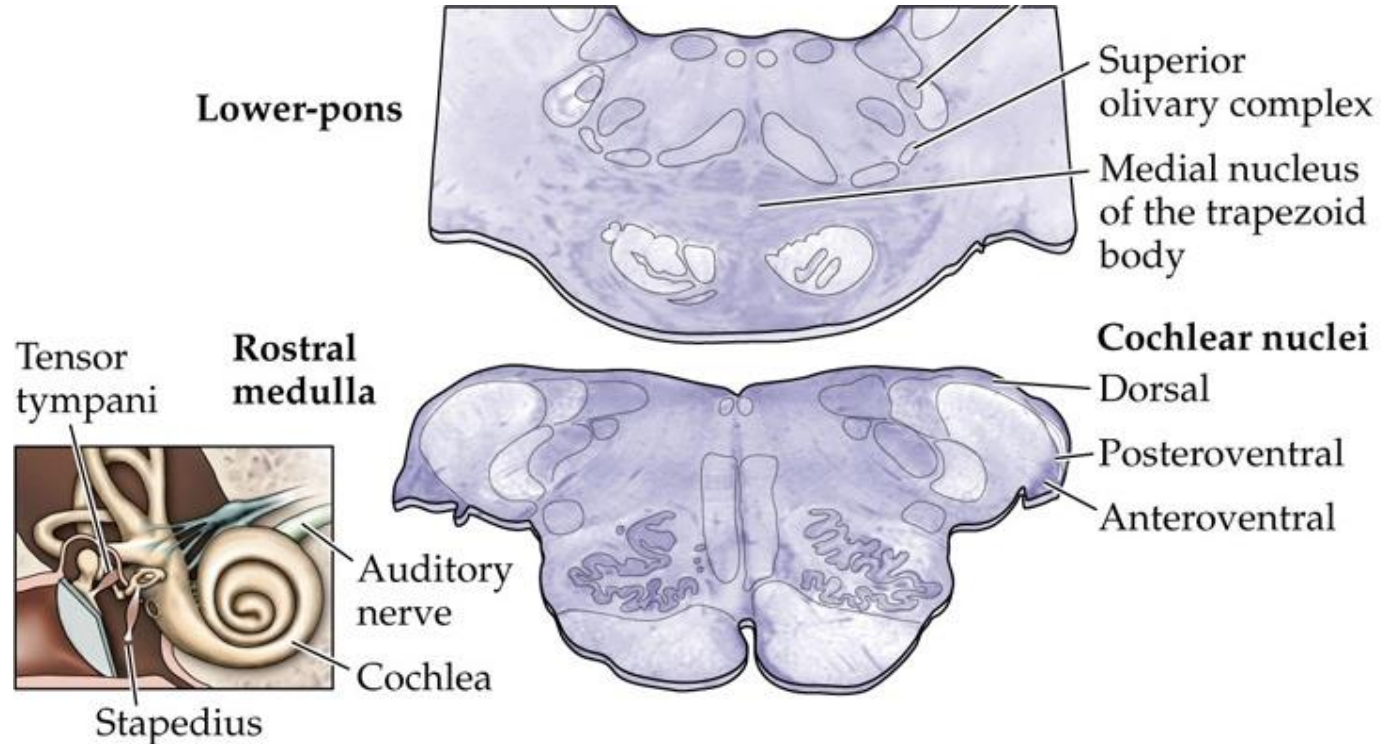
CONCEPT 10.5

Auditory pathways involve signals traveling bilaterally and in both feedforward and feedback directions



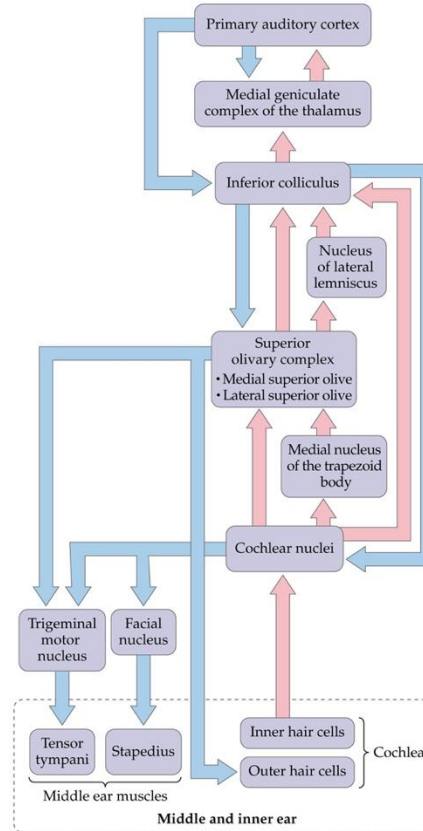
CONCEPT 10.5

Auditory pathways involve signals traveling bilaterally and in both feedforward and feedback directions



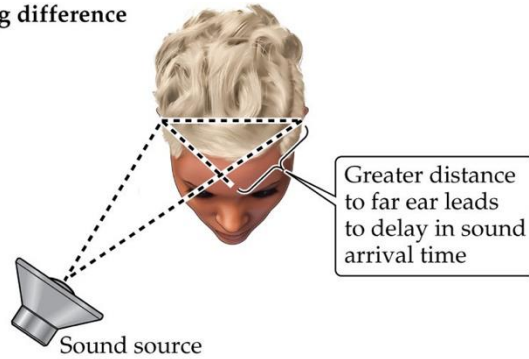
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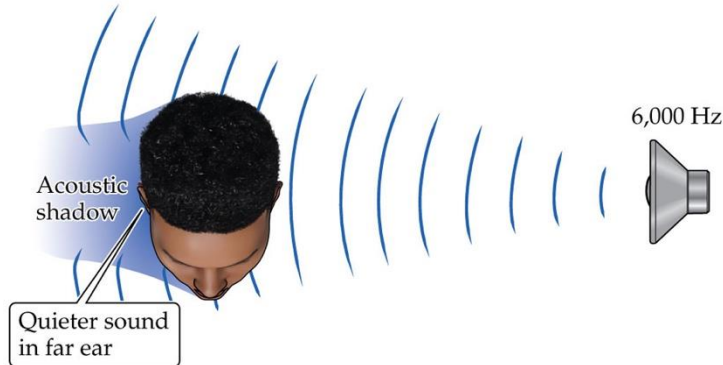


CONCEPT 10.8 Sounds are localized based on frequency, relative loudness, and relative time of arrival at the ears.

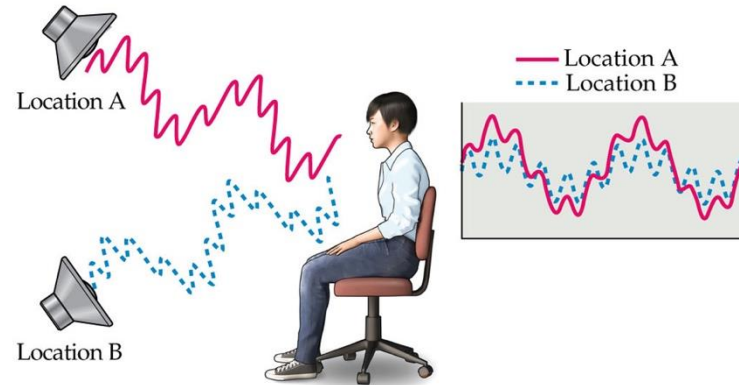
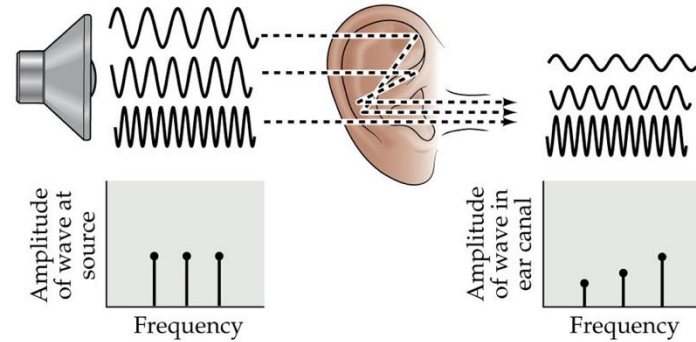
(A) Interaural timing difference



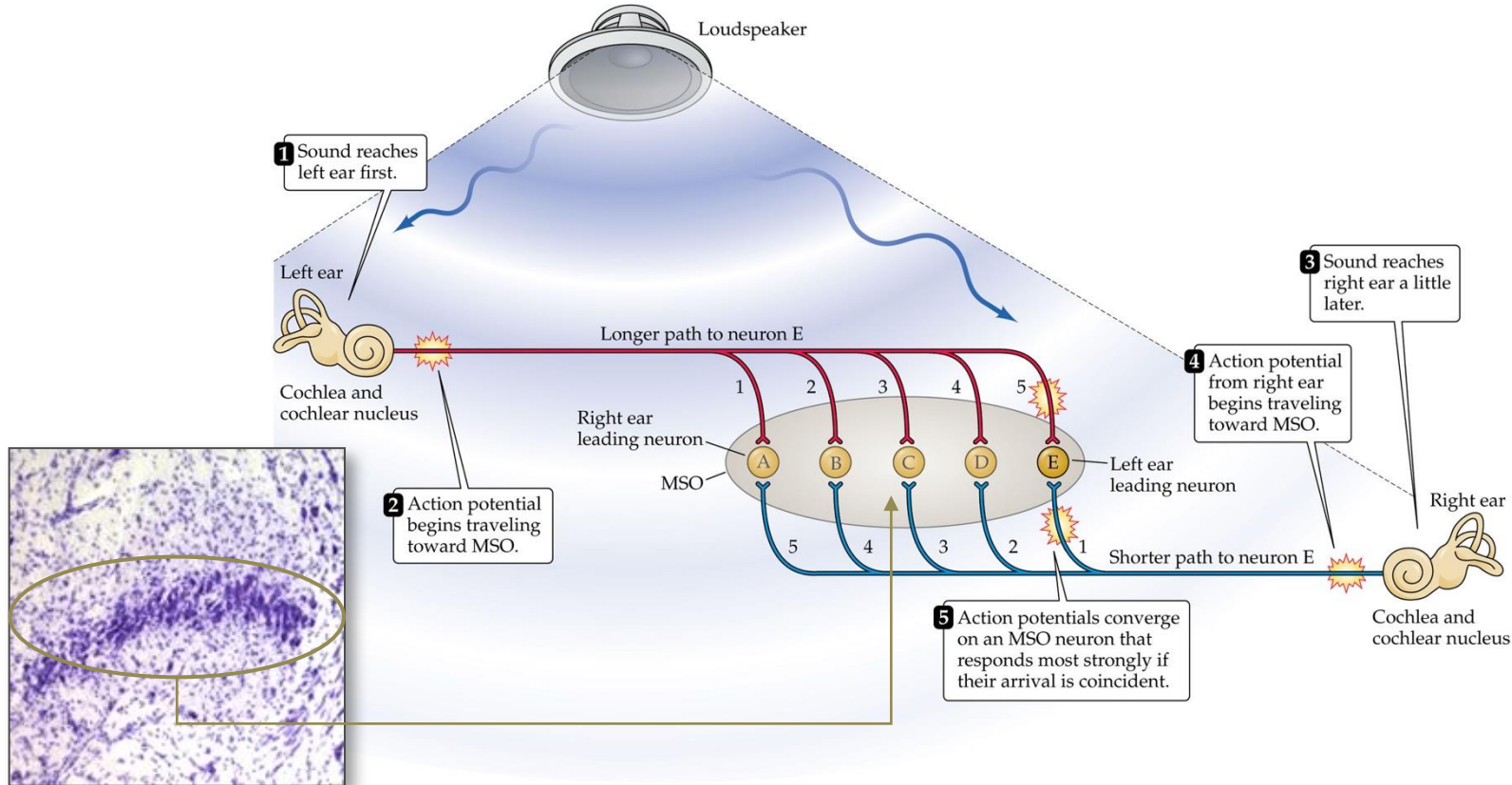
(B) Interaural level difference



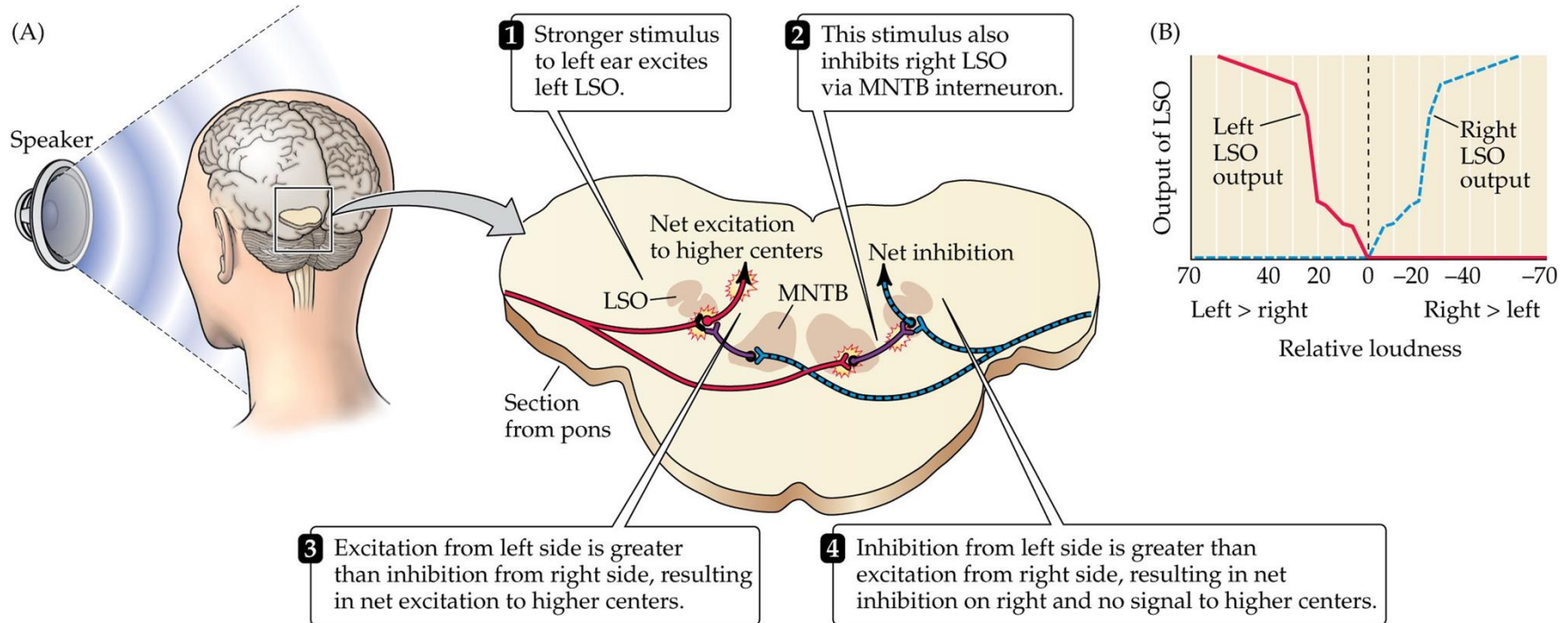
(C) Spectral cues



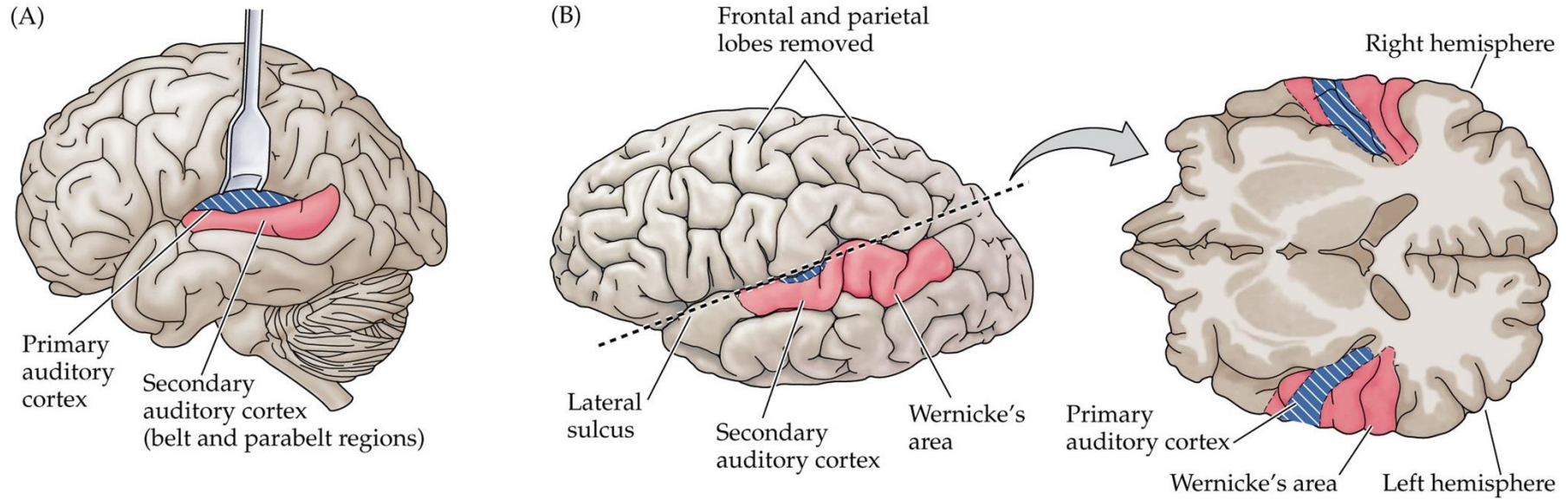
The MSO computes the location of a sound by interaural timing differences



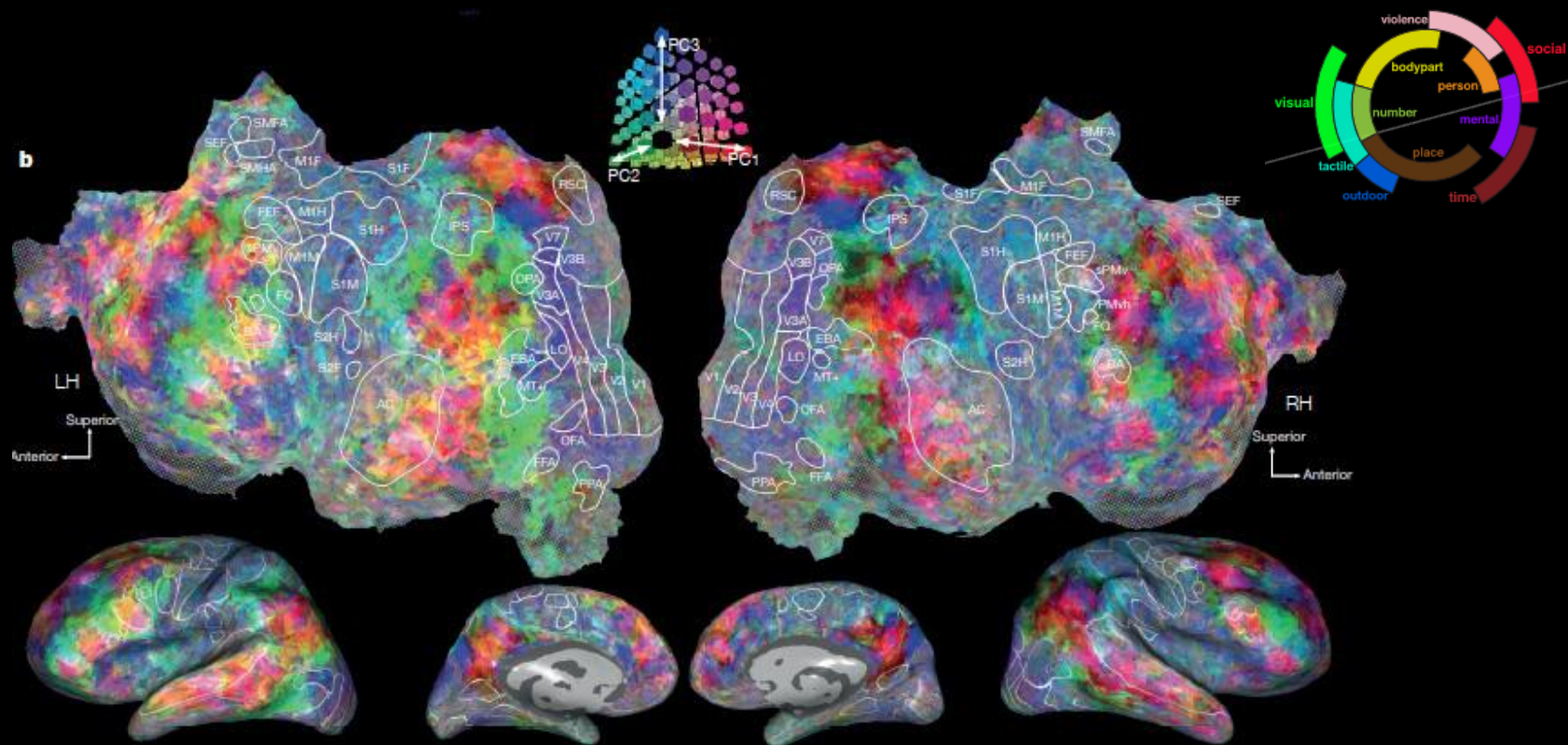
The LSO encodes sound location through interaural intensity differences



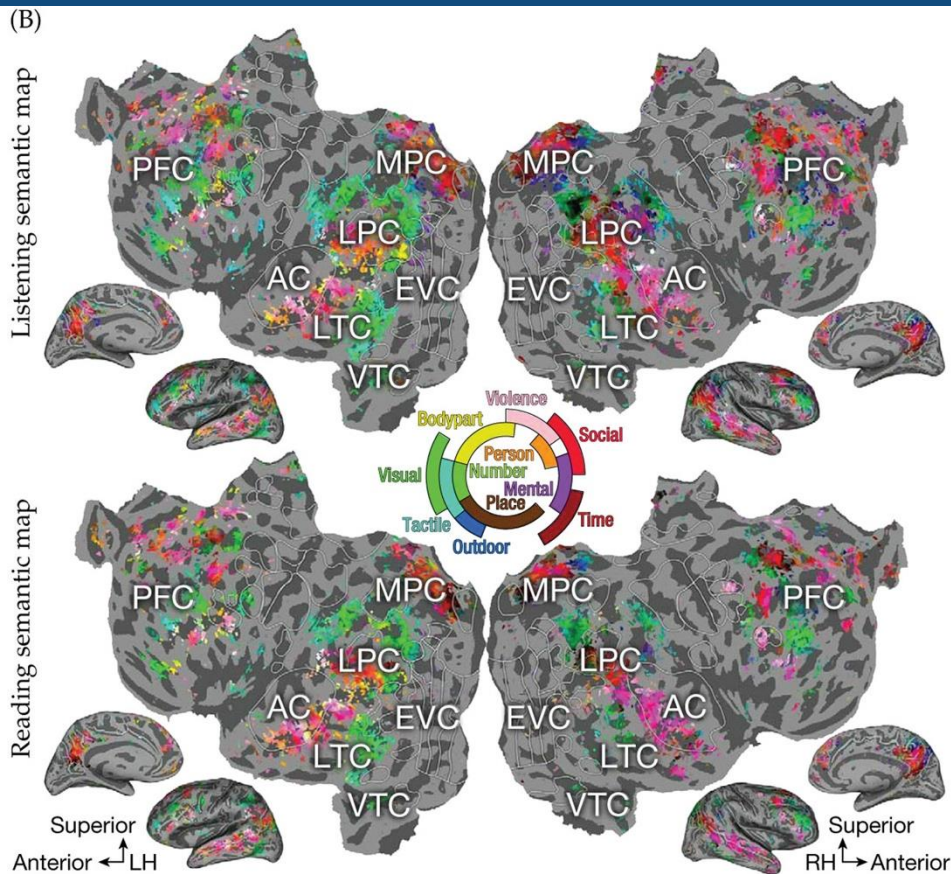
The auditory cortex comprises core (primary) and belt areas in the superior and lateral aspects of the temporal lobe



Semantic mapping in the human cerebral cortex

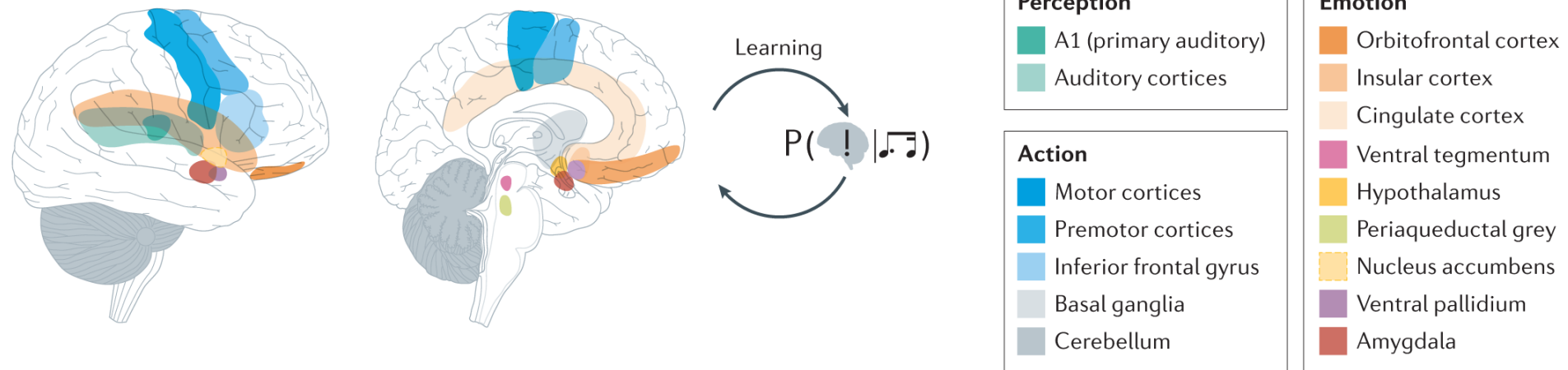


Listening and reading yield similar semantic maps



Making music is one of the most powerful modulators of brain function

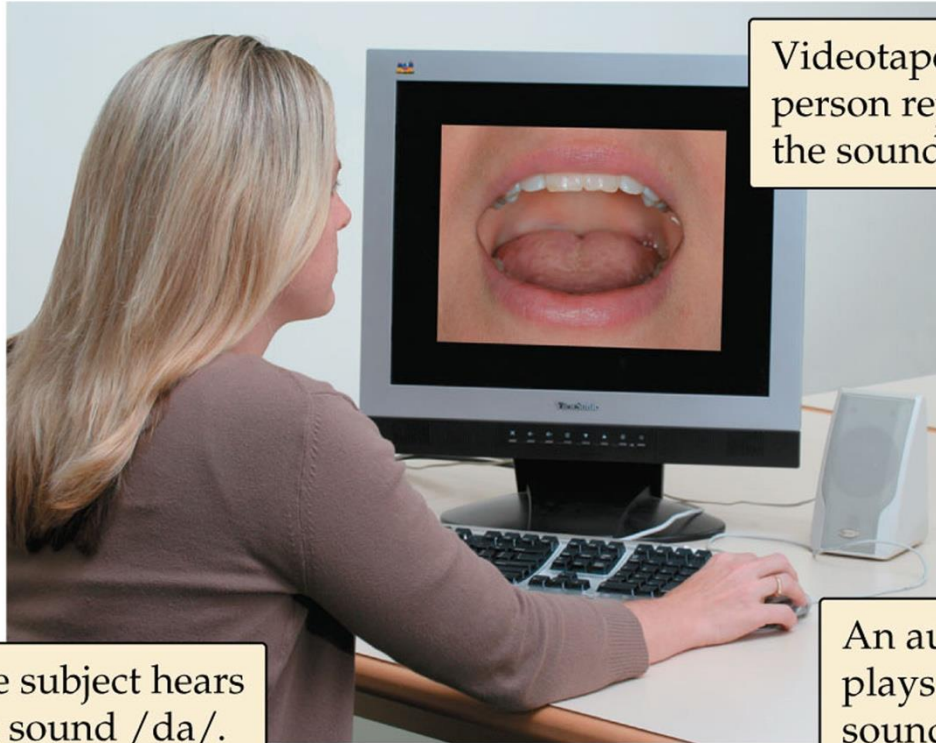
d Music brain networks



The McGurk effect illustrates visual-auditory association in language encoding

(A)

© Juniors Bildarchiv
GmbH/Alamy Stock Photo



Videotape shows a person repeating the sound /ga/.

The subject hears the sound /da/.

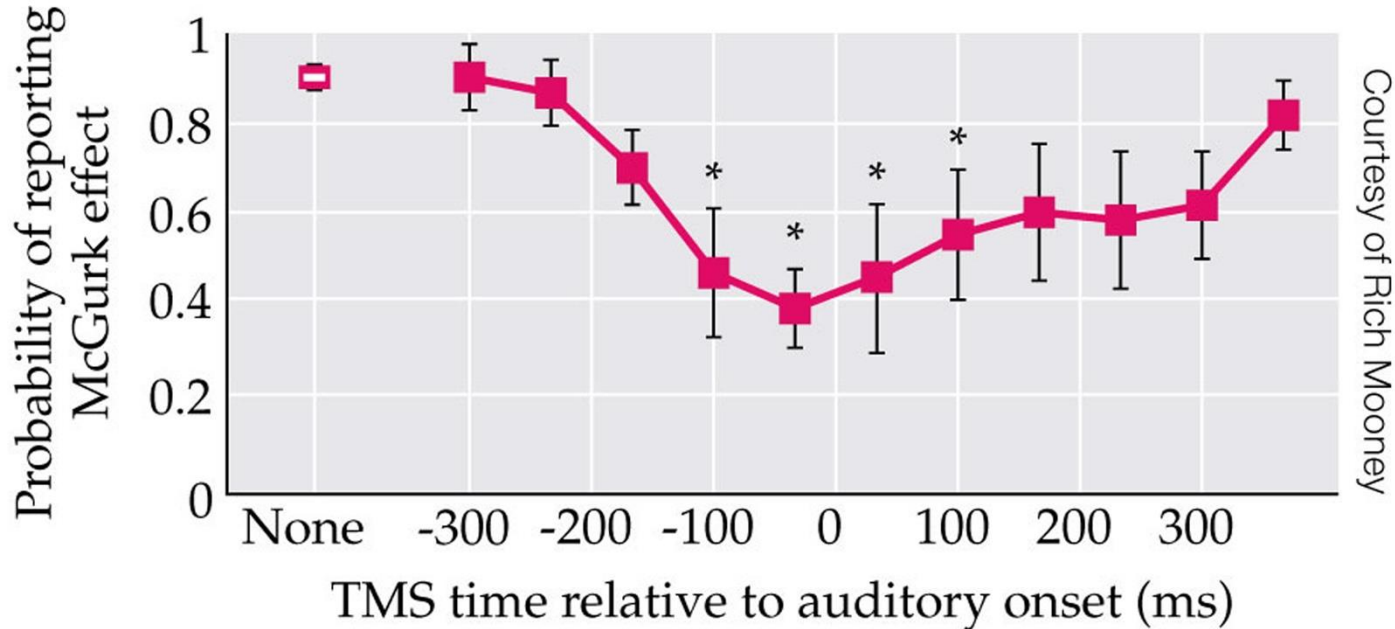
An audio track plays the sound /ba/.

[<https://www.youtube.com/watch?v=2k8fHR9jKVM>]

[Figure 31.5(A) from *Neuroscience*, 7th Ed., © 2023 Oxford University Press; see <https://www.youtube.com/watch?v=2k8fHR9jKVM>]

The McGurk effect can be disrupted with TMS applied to the lateral temporal region

(C)



CONCEPT 10.9 Hearing coordinates with vision via numerous signal transformations and robust intersensory cross-talk

